# Beam-Steered Single-LED Single-Photodiode Optical Wireless Positioning for 3D Indoor Localization

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***Abstract*—Indoor localization with radio-frequency signals often suffers from multipath, non-line-of-sight propagation, and spectrum congestion, limiting accuracy unless dense infrastructure is deployed. This paper introduces a beam–steered single-LED, single-photodiode (PD) optical wireless positioning (OWP) architecture that achieves full *3D* localization without receiver rotation, cameras, or PD arrays. The method follows a two-stage pipeline: (i) direction finding from $K$ steered transmitter orientations using power-ratio constraints, and (ii) range recovery from a single beam-aligned power measurement. We derive a composite Cramér–Rao lower bound (CRLB) and position-error bound (PEB) for the joint observation model, establish identifiability and Fisher information rank conditions as a function of $K$, and cast the steering-pattern design as a genetic algorithm that minimizes the PEB over a 3D testbed. For stage (i), we develop both a constrained nonlinear (NL) estimator and closed-form direction estimators: a statistically efficient generalized least squares (GLS) solution that accounts for heteroscedastic, correlated ratio errors, and a lightweight weighted least squares (WLS) approximation; all reduce inference to a single $3\times3$ eigenproblem (NL used for comparison). Simulations in a $3\times3\times2\,\mathrm{m}^3$ room show centimeter-level accuracy with modest $K$: optimized sets yield sub–$1.5\,\mathrm{cm}$ median PEB for $K\in[3,9]$; GLS attains $\mathrm{RMSE}=2.72\,\mathrm{cm}$ at $K=5$ and $2.41\,\mathrm{cm}$ at $K=9$, with an *average positioning error (APE) of* $1.54\,\mathrm{cm}$ *at* $K=9$, tracking the CRLB within $\sim1.75\times$ while offering $\mu$s-level latency. Concentrating complexity in a single steerable luminaire enables deployment with minimal user equipment and naturally supports multi-user time multiplexing.**

***Index Terms*—3D localization, Cramér–Rao lower bound (CRLB), generalized least squares (GLS), indoor localization, light emitting diode (LED), optical wireless positioning (OWP), received signal strength (RSS), single LED**

## I. Introduction

GLOBAL Navigation Satellite Systems (GNSS) provide ubiquitous outdoor positioning but are unreliable indoors due to severe attenuation, blockage, and multipath, which motivates dedicated indoor positioning systems (IPS) [1]. Among IPS, radio-frequency (RF) techniques leveraging existing communications infrastructure, such as Wi-Fi [2], Bluetooth Low Energy (BLE) beacons [3], ultra-wideband (UWB) [4], Radio Frequency Identification (RFID) [5], and emerging 5G positioning [6], are widely studied and deployed [6], [7]. However, in cluttered interiors they are strongly affected by multipath, non-line-of-sight (NLOS) conditions, and time-varying interference in shared spectrum, typically yielding meter-level accuracy unless dense infrastructure, calibration, or specialized hardware ~~is used~~ [8], [9].

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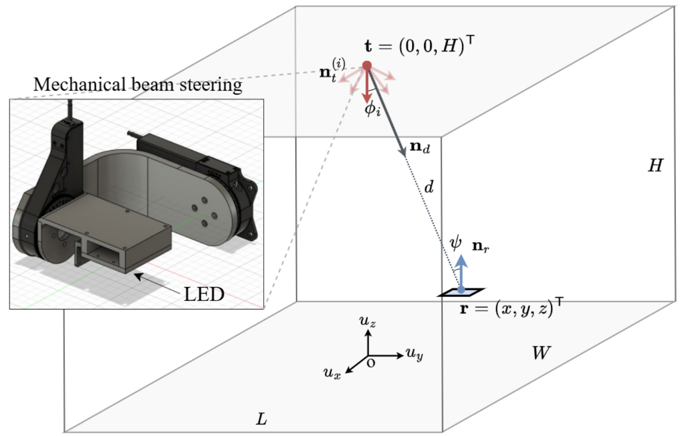


Fig. 1. Configuration of the system single-LED Optical Wireless Positioning.

In this context, Optical Wireless Positioning (OWP) leverages low-cost, small-form-factor light-emitting diodes (LEDs) that transmit modulated optical signals, often imperceptible to humans, captured by photodiodes (PDs) or cameras to transform lighting into a positioning network [10]. The spatial confinement of light ~~(no wall penetration)~~, immunity to RF electromagnetic interference, high spatial reuse, and more deterministic line-of-sight optical channels enable centimeter-level localization with cost-effective luminaires [11], making OWP attractive for navigation and asset tracking in malls, airports, and museums where GNSS is unavailable, as well as in electromagnetic-sensitive environments such as hospitals and airplane cabins [10]. Moreover, the same LED infrastructure can provide Optical Wireless Communication (OWC), often termed Li-Fi [12], so that illumination, data, and positioning coexist in a unified platform, albeit with practical constraints on illumination and LOS availability.

Position estimation in OWP follows two paradigms: data-driven and model-based. Data-driven methods include fingerprinting with k-nearest neighbors; support vector machines/regression; random forests and gradient boosting; Gaussian process regression; multilayer perceptrons; convolutional neural networks for camera or structured PD signals; recurrent/temporal models (LSTM/GRU/Transformers); and autoencoder- or metric-learning–based embeddings for domain adaptation [13], [14]. While effective with abundant labeled data, these approaches are often opaque, sensitive to domain shift (hardware, illumination, occlusion), and require periodic retraining, offering limited physical guarantees [15].

TABLE I
PERFORMANCE COMPARISON OF SINGLE-LED VLP SYSTEMS.

| Year | Work | Configuration | Rx Orientation | 2D/3D | Method | Error (APE) | Room ($m^3$) | Notes |
|---|---|---|---|---|---|---|---|---|
| 2020 | [17] | 1 LED / 4 PD (1 horizontal + 3 tilted) | Fixed | 3D | Model-based (Geometrical Equations) | 2.52 cm | $1 \times 1 \times 1.5$ | Sim. |
| 2021 | [18] | 1 LED / 3 tilted PD | Fixed | 2D | Long Short Term Memory- Fully Connected Network | 0.92 cm | $1 \times 1.1 \times 1.75$ | Exp. |
| 2021 | [19] | 1 LED / 4 PD (1 horizontal + 3 tilted) | Fixed | 2D | Bayesian algorithm | 9.27 cm | $3 \times 3 \times 3$ | Exp. |
| 2022 | [20] | 1 LED / 1 Rotatable PD | Fixed | 2D | Extreme Learning Machine | 1.74 cm | $4 \times 4 \times 3.1$ | Sim. |
| 2023 | [21] | 1 LED / 4 PD | Fixed | 2D | Least Square | 7 cm (CDF$_{90\%}$) | $6 \times 6 \times 3$ | Exp. |
| 2024 | [22] | 1 reorientable LED / 1 PD | Arbitrary | 2D | Model-based (Power Ratio) | 5.9 cm (CDF$_{90\%}$) | $4 \times 4 \times 2.5$ | Exp. |
| 2024 | [23] | 1 LED / 4 PD | Fixed | 2D | Extreme Learning Machine / WKNN | 2.93 cm | $5 \times 5 \times 3$ | Sim. |
| 2024 | [24] | 1 LED / 4 PD (1 horizontal + 3 tilted) | Fixed | 3D | Deep Residual Shrinkage method | 5.85 cm | $3.6 \times 3.6 \times 3.0$ | Exp. |
| 2024 | [25] | 1 LED / 1 PD (Single-Tilted-Rotatable PD) | Rotatable | 2D | Least Squares + Improved Grey Wolf Optimizer (LS-IGWO) | 1.65 cm | $0.5 \times 0.5 \times 0.3$ | Exp. |
| 2024 | [26] | 1 LED / 4 PD + IMU (orientation) | Arbitrary | 3D | MLP | 1.77 cm | $1.5 \times 1.5 \times 1.5$ | Exp. |
| 2025 | [27] | 1 LED / 1 Rotatable PD | Fixed | 2D | Least Square | 4.96 cm | $5 \times 5 \times 4$ | Exp. |
| 2025 | [28] | 1 LED / 16 IRS / 1 PD | Fixed | 2D | Levenberg–Marquardt algorithm | 5.6 cm (RMSE) | $4 \times 5 \times 5$ | Sim. |
| 2025 | Ours | 1 reorientable LED / 1 PD | Arbitrary | 3D | GLS | 1.54 cm | $3 \times 3 \times 2$ | Sim. |

*Note:* “Sim.” denotes simulation results; “Exp.” denotes experimental results.

By contrast, model-based estimators grounded in radiometry and geometry (e.g., Lambertian propagation, field-of-view constraints, pose kinematics) provide interpretability, sample efficiency, analyzable uncertainty, and reliable extrapolation, and they support joint system co-design [16].

Most existing OWP implementations rely on traditional multi-LED transmitter constellations, where multiple fixed LEDs with known positions act as anchors [29]. Using multiple LEDs allows the receiver to compute its position via triangulation, trilateration, or multilateration, analogous to how GNSS uses many satellites [10]. However, this multi-LED requirement complicates practical deployment. It presumes a dense infrastructure; at least four LEDs must be concurrently visible to the receiver to resolve a 3D position [30]–[33], which may not be true in environments with only sparse lighting. Installing and maintaining a dense LED network, with unique IDs or modulation for each lamp, increases installation cost and complexity [34]. Consequently, reducing the infrastructure burden by localizing with single light sources is a highly desirable goal in the field [13].

Recent research has explored OWP schemes that break from the multi-LED paradigm. One notable direction is single-LED positioning, which aims to achieve localization with only one transmitter [17]–[28]. A single light source dramatically simplifies infrastructure and calibration [35]. However, a lone transmitter cannot provide full positional information at a fixed orientation, since fundamentally one measurement is insufficient to resolve a 3D position. To overcome this, single-LED OWP systems incorporate additional cues or mechanisms in the UE side. Some approaches use additional sensors or diversity at the receiver to glean more information from the single source. While these single-LED solutions eliminate the need for multiple transmitters, they introduce their own complexities in the UE side, requiring user participation (device rotation), specialized receiver hardware (camera or multi-PD array), or additional aiding sensors like inertial measurement units. Moreover, some early single-LED demonstrations focused only on 2D positioning, assuming the receiver was on a fixed plane or height, which limits applicability. These limitations motivate the search for a new 3D single-LED positioning framework.

In parallel, recent advances in OWC have delivered agile beam-steering transmitters, ranging from high-speed MEMS micromirror gimbals [36], through liquid-crystal or metasurface deflectors that widen the field-of-view without sacrificing speed [37], to compact silicon-photonics optical-phased arrays that shift their beam tens of degrees in micro-seconds with no moving parts [38]. These demonstrations confirm that equipping a single indoor luminaire with a steering mechanism is feasible and highly synergistic.

In this work, we study a novel beam–steered single-LED single-PD OWP architecture for 3D indoor localization. The approach follows a two-stage pipeline—(i) direction finding from $K$ steered orientations, and (ii) range recovery using a high-SNR beam-aligned measurement—without requiring receiver rotation or PD arrays. The main contributions are summarized as follows:

- A novel single-LED single-PD 3D OWP framework that achieves full 3D localization without receiver rotation, camera sensing, or PD arrays, enabled by beam steering.
- CRLB/PEB analysis for a composite observation model. We derive Fisher-information expressions and the associated position-error bound (PEB) for the joint set of $K$ orientation-averaged received powers plus one beam-aligned range statistic, and we characterize identifiability and FIM rank conditions as a function of $K$.
- GA-based orientation-set design. We cast the transmitter orientation selection as a genetic-algorithm optimization that minimizes the RMSE PEB over a 3D testbed, yielding practical design rules for the choice of $K$ and set of orientations.
- Closed-form linear direction estimators. Using a ratio-based linearization, we develop a statistically efficient GLS estimator that accounts for heteroscedastic and correlated ratio errors, alongside a lightweight WLS approximation suitable for low-power implementations.

This paper is organized as follows. Section II presents the system model and the two-stage 3D localization procedure. Section III derives the position-error bound (PEB) for single-LED positioning. Section IV optimizes the transmitter beam-orientation set. Section V develops a nonlinear (NL) direction estimator. Section VI introduces closed-form linear estimators (GLS and WLS). Section ?? reports simulation results. Section VIII concludes the paper.

*Notation:* Bold lowercase letters denote column vectors (e.g., $\mathbf{x}$) and bold uppercase letters denote matrices (e.g., $\mathbf{A}$); scalars are in lightface. The Euclidean norm is $\|\cdot\|$, the inner product is $\mathbf{a}\cdot\mathbf{b}$, the trace is $\mathrm{tr}(\cdot)$, and the gradient with respect to $\mathbf{r}$ is $\nabla_{\mathbf{r}} = [\partial/\partial x, \partial/\partial y, \partial/\partial z]^{\mathsf{T}}$. The transpose is $(\cdot)^{\mathsf{T}}$, the identity and zero matrices are $\mathbf{I}_n$ and $\mathbf{0}$, respectively, and $\mathrm{diag}(\cdot)$ forms a diagonal matrix from its arguments. The smallest eigenvalue and a corresponding unit-norm eigenvector of a symmetric matrix $\mathbf{M}$ are denoted by $\lambda_{\min}(\mathbf{M})$ and $\mathbf{v}_{\min}(\mathbf{M})$; the Loewner order $\succeq$ indicates positive semidefiniteness. Calligraphic symbols (e.g., $\mathcal{R}$) denote sets and $|\mathcal{A}|$ their cardinality; $\mathbb{R}^n$ is the $n$-dimensional real space and $\mathbb{S}^2$ the unit sphere in $\mathbb{R}^3$. Random variables follow the convention $x \sim \mathcal{N}(\mu, \sigma^2)$, with expectation $\mathbb{E}[\cdot]$ and variance $\mathrm{Var}[\cdot]$. A hat $\hat{(\cdot)}$ denotes an estimate and an overbar $\bar{(\cdot)}$ a sample mean. ~~The natural logarithm is $\ln(\cdot)$ and the base-10 logarithm is $\log_{10}(\cdot)$; signal-to-noise ratios in decibels use $10\log_{10}(\cdot)$.~~

## II. System Model and Proposed 3D Localization Method

### A. System Model

Fig. 1 illustrates the single-LED single-PD OWP setup analyzed in this work. A LED transmitter is mounted on the ceiling at a known position $\mathbf{t} = [0, 0, H]^{\mathsf{T}}$ and can steer its optical axis along a unit vector $\mathbf{n}_t^{(i)} \in \mathbb{S}^2$ for the $i$-th orientation, $i = 1, \ldots, K$. The receiver is a single PD with fixed normal along the vertical axis $\mathbf{n}_r = [0, 0, 1]^{\mathsf{T}}$ located at an unknown position $\mathbf{r} = [x, y, z]^{\mathsf{T}}$. Let $\mathbf{d} = \mathbf{r} - \mathbf{t}$ denote the transmitter–receiver displacement vector, $d = \|\mathbf{d}\|_2$ the separation distance, and $\mathbf{n}_d = \mathbf{d}/d$ the corresponding unit direction vector.

For the $i$-th LED orientation, the received signal strength (RSS) at the PD is modeled as:

$$S_r^{(i)} = R_{\mathrm{pd}} P_t \, h_{\mathrm{LOS}}^{(i)} + w_i, \tag{1}$$

where $R_{\mathrm{pd}}$ is the PD responsivity and $w_i \sim \mathcal{N}(0, \sigma_w^2)$ models a zero-mean additive white Gaussian noise (AWGN) term arising primarily from shot and Johnson noise, and assumed independent and identically distributed (i.i.d.) across orientations. Dividing (1) by $R_{\mathrm{pd}}$ yields the received optical power:

$$P_r^{(i)} = P_t \, h_{\mathrm{LOS}}^{(i)} + n_i, \tag{2}$$

where $P_t$ is the transmitted radiant power, $n_i \sim \mathcal{N}(0, \sigma^2)$ is zero-mean AWGN, remains i.i.d., with variance $\sigma^2 = (\sigma_w^2/R_{\mathrm{pd}}^2)$ , and $h_{\mathrm{LOS}}^{(i)}$ denotes the line-of-sight (LOS) channel gain, which for a Lambertian source is given by [39]:

$$h_{\mathrm{LOS}}^{(i)} = \begin{cases} \dfrac{(m+1)A_{\mathrm{det}}}{2\pi d^2} \cos^m(\phi_i)\cos(\psi), & \psi \le \Psi_{\mathrm{FOV}}, \\ 0, & \text{otherwise,} \end{cases} \tag{3}$$

where $m$ is the Lambertian order of the LED, related to the semi-angle at half power $\Phi_{1/2}$ by $m = -\ln 2/\ln(\cos(\Phi_{1/2}))$, $A_{\mathrm{det}}$ the photodiode effective area, $\Psi_{\mathrm{FOV}}$ the receiver field of view, and the irradiance/incidence cosines geometrically given by:

$$\cos\phi_i = \frac{\mathbf{n}_t^{(i)} \cdot \mathbf{d}}{d}, \qquad \cos\psi = \frac{\mathbf{n}_r \cdot (-\mathbf{d})}{d}. \tag{4}$$

At every orientation $i$ the receiver acquires and averages $N$ samples, producing the unbiased estimator of the received power:

$$\bar{P}_r^{(i)} = \frac{1}{N}\sum_{k=1}^{N} P_{r,k}^{(i)} \tag{5}$$

The corresponding instantaneous signal-to-noise ratio (SNR) and its testbed average are:

$$\mathrm{SNR}_r^{(i)} = \frac{(R_{\mathrm{pd}} P_t \, h_{\mathrm{LOS}}^{(i)})^2}{\sigma_w^2}, \tag{6a}$$

$$\overline{\mathrm{SNR}}_r = \frac{1}{K}\sum_{i=1}^{K} \mathrm{SNR}_r^{(i)}, \tag{6b}$$

$$\overline{\mathrm{SNR}} = \frac{1}{|\mathcal{R}|}\sum_{r \in \mathcal{R}} \overline{\mathrm{SNR}}_r. \tag{6c}$$

where $\mathcal{R}$ denotes the set of receiver testbed positions. Results are reported in decibels through $10\log_{10}\big(\overline{\mathrm{SNR}}\big)$.

### B. General 3D single-LED Positioning Procedure

The proposed localisation pipeline comprises two consecutive stages.

*1) Direction finding:* The first stage estimates the direction vector $\mathbf{n}_d$ by steering the LED through the $K$ predefined orientations $\{\mathbf{n}_t^{(i)}\}_{i=1}^{K}$ and processing the power means $\{\bar{P}_r^{(i)}\}_{i=1}^{K}$. Section V (NL estimator) and Section VI (linear GLS/WLS estimators) detail statistically grounded solutions.

*2) Distance recovery:* After direction finding, the LED is aligned with the estimated direction $\hat{\mathbf{n}}_d$ and, for beam-forming symmetry, the receiver is re-oriented to $\mathbf{n}_r = -\hat{\mathbf{n}}_d$. Under this beam-steered configuration and using (2), (3), and (4), the received power obeys:

$$P_r^* = \frac{C}{d^2} + n, \tag{7}$$

where $C$ is an optical constant calibrated once per hardware set-up:

$$C = \frac{P_t(m+1)A_{\mathrm{det}}}{2\pi}, \tag{8}$$

Collecting $N$ independent measurements $\{P_{r,j}^*\}_{j=1}^{N}$ and using (5) yields the Minimum Variance Unbiased (MVU) estimate $\bar{P}_r^* \sim \mathcal{N}\big(\mu^*, \sigma^2/N\big)$. Substituting $\bar{P}_r^*$ into (7) provides the distance estimate:

$$\hat{d} = \sqrt{\frac{C}{\bar{P}_r^*}}, \tag{9}$$

and, finally, the 3D position estimate:

$$\hat{\mathbf{r}} = \mathbf{t} + \hat{d}\,\hat{\mathbf{n}}_d. \tag{10}$$

## III. Position Error Bound

This section derives the position error bound (PEB), i.e., the Cramér–Rao lower bound (CRLB) specialised to the receiver's 3D coordinate vector $\mathbf{r} = [x, y, z]^\mathsf{T}$. The PEB quantifies the minimum achievable variance of any unbiased estimator of $\mathbf{r}$ under the proposed OWP model and serves both as a benchmark for estimator performance and as an objective for system design.

### A. Joint statistical model of the observations

According to Section II, the sample-mean optical power, as defined in (5), for each of the $K$ transmitter orientations used during the direction finding stage is distributed as:

$$\bar{P}_r^{(i)} \sim \mathcal{N}\big(\mu_i(\mathbf{r}),\, \sigma^2/N\big), \quad i = 1, \dots, K, \tag{11}$$

where $\mu_i(\mathbf{r}) = P_t\, h_{\mathrm{LOS}}^{(i)}(\mathbf{r})$ and $h_{\mathrm{LOS}}^{(i)}$ is defined in (3) with the cosines in (4). During the distance recovery stage, the mean optical power in aligned configuration obeys:

$$\bar{P}_r^{*} \sim \mathcal{N}\big(\mu_{K+1}(\mathbf{r}),\, \sigma^2/N\big), \tag{12}$$

where $\mu_{K+1}(\mathbf{r}) = C/d^2(\mathbf{r})$ , $d(\mathbf{r}) = \|\mathbf{r} - \mathbf{t}\|$ and $C$ is the optical constant introduced in (8).

Stacking these $K + 1$ independent observations into $\mathbf{z} = [\, \bar{P}_r^{(1)}, \dots, \bar{P}_r^{(K)}, \bar{P}_r^{*}\,]^\mathsf{T}$ yields the multivariate Gaussian model:

$$\mathbf{z} \sim \mathcal{N}\big(\boldsymbol{\mu}(\mathbf{r}),\, (\sigma^2/N)\, \mathbf{I}_{K+1}\big), \tag{13}$$

with mean vector $\boldsymbol{\mu}(\mathbf{r}) = [\, \mu_1(\mathbf{r}), \dots, \mu_{K+1}(\mathbf{r})\,]^\mathsf{T}$ and diagonal covariance $\sigma^2/N$.

### B. Log-likelihood and score function

Under the assumption of independent Gaussian measurements, the log-likelihood of $\mathbf{r}$ given $\mathbf{z}$ is:

$$\ell(\mathbf{r}) = \ln L(\mathbf{r} \mid \mathbf{z}) = -\frac{N}{2\sigma^2} \sum_{i=1}^{K+1} \big[z_i - \mu_i(\mathbf{r})\big]^2 + c_0, \tag{14}$$

where $c_0$ is an additive constant independent of $\mathbf{r}$. Differentiating $\ell(\mathbf{r})$ with respect to $\mathbf{r}$ defines the score vector:

$$\mathbf{s}(\mathbf{r}) = \nabla_{\mathbf{r}}\, \ell(\mathbf{r}) = \frac{N}{\sigma^2} \sum_{i=1}^{K+1} \big[z_i - \mu_i(\mathbf{r})\big]\, \nabla_{\mathbf{r}}\, \mu_i(\mathbf{r}), \tag{15}$$

where $\nabla_{\mathbf{r}} = [\partial/\partial x, \partial/\partial y, \partial/\partial z]^\mathsf{T}$ is the gradient operator.

### C. Fisher Information Matrix

For distributions in the exponential family, such as the homoscedastic Gaussian model adopted here, the Fisher Information Matrix (FIM) can be obtained as the variance of the score vector, avoiding second-order differentiation and yielding greater numerical stability [40]:

$$\mathbf{I}(\mathbf{r}) = \mathrm{Var}\big[\mathbf{s}(\mathbf{r})\big] = \mathbb{E}_{\mathbf{z}}\big[\mathbf{s}(\mathbf{r})\, \mathbf{s}(\mathbf{r})^\mathsf{T}\big] \tag{16}$$

$$= \frac{N}{\sigma^2} \sum_{i=1}^{K+1} \big[\nabla_{\mathbf{r}}\mu_i(\mathbf{r})\big]\, \big[\nabla_{\mathbf{r}}\mu_i(\mathbf{r})\big]^\mathsf{T}. \tag{17}$$

The second equality follows from $\mathbb{E}[z_i] = \mu_i(\mathbf{r})$, which implies $\mathbb{E}\big[(z_i - \mu_i)(z_j - \mu_j)\big] = 0$ for $i \neq j$. Hence the total information is the coherent superposition of the differential contributions from each LED orientation and the range measurement. Required gradients are:

*1) Direction-finding measurements:* For the $i$-th orientation:

$$\nabla_{\mathbf{r}}\mu_i(\mathbf{r}) = P_t\, \nabla_{\mathbf{r}} h_{\mathrm{LOS}}^{(i)}(\mathbf{r}), \tag{18}$$

which, by differentiating (3) via the chain rule on $d(\mathbf{r})$, $\cos\phi_i$ and $\cos\psi$, yields the closed-form

$$\nabla_{\mathbf{r}}\mu_i(\mathbf{r}) = \frac{C}{d^3} \Big[ m\, \cos^{m-1}\phi_i\, \cos\psi\, \mathbf{n}_t^{(i)} \,-\, \cos^m\phi_i\, \mathbf{n}_r \\ -\, (m+3)\, \cos^m\phi_i\, \cos\psi\, \mathbf{n}_d \Big], \tag{19}$$

which vanishes whenever $\psi > \Psi_{\mathrm{FOV}}$ since $h_{\mathrm{LOS}}^{(i)} = 0$ outside the receiver's field of view.

*2) Distance-recovery measurement:*

$$\nabla_{\mathbf{r}}\mu_{K+1}(\mathbf{r}) = -\frac{2C}{d^4}\big(\mathbf{r} - \mathbf{t}\big) = -\frac{2C}{d^3}\, \mathbf{n}_d, \tag{20}$$

where $\mathbf{n}_d$ is the unit direction vector defined in Section II.

### D. Formal definition of the PEB

Let $\hat{\mathbf{r}}$ be any unbiased estimator of $\mathbf{r}$. The Cramér–Rao inequality asserts:

$$\mathrm{Cov}\big[\hat{\mathbf{r}}\big] \succeq \mathbf{I}^{-1}(\mathbf{r}). \tag{21}$$

Accordingly, a scalar performance metric, the PEB, is defined as:

$$\mathrm{PEB}(\mathbf{r}) = \sqrt{\mathrm{tr}\big\{\mathbf{I}^{-1}(\mathbf{r})\big\}}. \tag{22}$$

The square root of the trace of $\mathbf{I}^{-1}$ corresponds to the minimum achievable root-mean-square error (RMSE) in position estimation for the described OWP configuration. A small PEB indicates that the chosen set of LED orientations and the range measurement provide sufficient and complementary information for high-precision localisation; conversely, a large PEB suggests the need to increase $K$, improve the SNR, or redesign the orientation pattern.

### E. Number of LED Orientations

*1) Underdetermined regime ($K \leq 2$):* In this regime, the Fisher information matrix $\mathbf{I}(\mathbf{r})$ from (17) is rank-deficient. Each of the $K$ orientation measurements contributes at most to one independent gradient vector, and the distance-recovery measurement adds a single additional vector. When $K \leq 2$, there are fewer than three linearly independent gradients in total, so $\mathbf{I}(\mathbf{r})$ is singular and the CRLB on any unbiased position estimate diverges [29].

*2) Just-determined regime ($K = 3$):* With three distinct, non-coplanar LED orientations plus the range measurement, the total of four gradient vectors can span the 3D parameter space. In generic geometries the FIM is full rank and $\mathbf{I}^{-1}(\mathbf{r})$ exists, yielding a finite PEB. However, the absence of redundancy makes the bound highly sensitive to small perturbations in orientation geometry [18].

*3) Overdetermined regime ($K \geq 4$):* For four or more orientations, the FIM accumulates redundant outer- products of gradients. This overdetermination improves the numerical conditioning of $\mathbf{I}(\mathbf{r})$ and generally lowers the PEB, since extra measurements provide additional independent information beyond the minimal requirement [10], [26].

## IV. Optimization of the Orientation Set

Equation (19) shows that the FIM $\mathbf{I}(\mathbf{r})$ depends explicitly on the transmitter orientation vectors $\{\mathbf{n}_t^{(i)}\}_{i=1}^K$, and hence so does the PEB in (22). Accordingly, selecting the orientation set that minimises the PEB is essential for optimal localisation accuracy. Since $\mathrm{PEB}(\{\mathbf{n}_t^{(i)}\})$ is non-convex in the orientation parameters, direct analytical minimisation is intractable. To address this, we employ a genetic algorithm (GA) to search for the orientation ensemble that minimises the global PEB for a given $K$.

TABLE II
Parameters used for GA optimization.

| System Parameters | |
|---|---|
| **Parameter** | **Value** |
| Transmitter position | $T = (0, 0, 2)$ m |
| Transmitted optical power | $P_t = 0.405$ W |
| Half-power semi-angle | $\Phi_{1/2} \in \{30°, 45°, 60°\}$ |
| Lambertian order | $m = -\ln 2/\ln(\cos(\Phi_{1/2}))$ |
| Photodiode effective area | $A_{\text{det}} = 4.8 \times 5.5\,\text{mm}^2$ |
| Receiver field of view (FOV) | $\Psi_{\text{FOV}} = 85°$ |
| Electrical noise variance per sample | $\sigma_w^2 = 1.19 \times 10^{-14}\,\text{A}^2$ |
| PD responsivity | $R_{\text{pd}} = 0.63$ A/W |
| Noise variance per sample | $\sigma^2 = 3 \times 10^{-14}\,\text{W}^2$ |
| Samples per orientation | $N = 1000$ |
| **Genetic Algorithm** | |
| **Parameter** | **Value** |
| Initial population | 300 |
| Maximum generations | 200 |
| Crossover rate | 0.8 |
| Search range | $\theta_i \in [0, 80°],\ \rho_i \in [0, 360°]$ |
| Optimization metric | RMSE |
| Decision variables | $[\theta_1, \rho_1, \ldots, \theta_K, \rho_K]$ |
| **3D testbed** | |
| **Parameter** | **Value** |
| Room dimensions (L×W×H) | $3 \times 3 \times 2\,\text{m}^3$ |
| Range in $x, y$ | $[-1.5,\ 1.5]$ m |
| Range in $z$ | $[0,\ 1.2]$ m |
| Grid step | 0.2 m |
| Total number of points tested | 1792 |

Each GA individual encodes $K$ steering directions via tilt–azimuth pairs $\{(\theta_i, \rho_i)\}_{i=1}^K$, yielding a $2K$-dimensional genotype $\mathbf{x} = [\theta_1, \rho_1, \ldots, \theta_K, \rho_K]^\mathsf{T}$. We adopt a nadir-referenced spherical parameterisation where $\theta_i$ is the tilt from $-\mathbf{u}_z$ and $\rho_i$ is the azimuth in the $x$–$y$ plane measured counterclockwise from $+\mathbf{u}_x$, with domains $\theta_i \in [0, \theta_{\max}]$ and $\rho_i \in [-0°, 360°]$. The mapping to unit-norm orientation vectors in the reference frame $(\mathbf{u}_x, \mathbf{u}_y, \mathbf{u}_z)$ is:

$$\mathbf{n}_t^{(i)} = \big(\sin\theta_i \cos\rho_i,\ \sin\theta_i \sin\rho_i,\ -\cos\theta_i\big). \tag{23}$$

Spherical coordinates are preferred to Cartesian because they describe unit-norm directions with two angles, avoiding the nonlinear unit-norm constraint and providing bounded domains that simplify GA search.

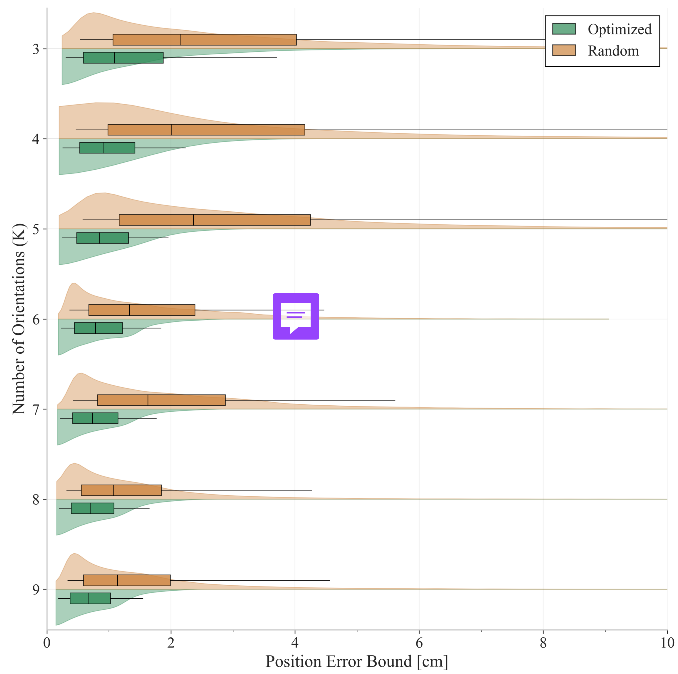


Fig. 2. RMSE performance in a $3 \times 3$m testbed using an optimal set of $K$ orientations versus a sub-optimal set of $K$ orientations.

The GA's fitness metric is defined as the RMSE of the PEB over all 3D grid points spanning the testbed, computed by evaluating $\mathrm{PEB}(\mathbf{r})$ from (22). Standard genetic operators such as tournament selection, crossover probability $p_c$, and mutation rate $p_m$, are applied over $G$ generations. The algorithm is implemented in MATLAB; optimization parameters are summarised in Table II.

Table III lists the optimal $K$-orientation sets for the system parameters in Table II. Each set is unique and no set is a subset of another. For every optimal orientation ensemble, we compute the PEB (or equivalently the RMSE error) at each receiver location within the testbed grid. In Fig. 2, the box-and-whisker plots for the optimal sets are shown in green, alongside those for randomly chosen orientations. Quantitatively, the optimized sets achieve a median of PEB below 1.5 cm for all $K \in \{3, \ldots, 9\}$ (worst median 1.43 cm at $K = 3$) and reach 0.74 cm at $K = 9$ (a 49%–72% reduction relative to random). These gains are reflected in the distributions, where the optimized sets exhibit lower medians and interquartile ranges, with the central tendency improving monotonically as $K$ increases.

Figure 3 presents spatial heat maps of the PEB across a $3 \times 3$ m floor at $z = 0.8$ m for $K = 5$ orientations with the transmitter at the room centre. In the optimal configuration (Fig. 3a), the highest errors occur uniformly along the boundaries and especially at the corners, analogous to multi-transmitter layouts, but remain spatially consistent. By contrast, the random configuration (Fig. 3b) exhibits both larger and more irregular error patterns.

Figure 4 depicts the 90th-percentile of the PEB distribution ($\mathrm{PEB}_{90\%}$) for the optimal $K$-orientation set under three $\Phi_{1/2}$ scenarios at an effective SNR of 14 dB. In all scenarios,

TABLE III
OPTIMAL ORIENTATIONS $(\theta_i, \rho_i)$ IN DEGREES [°] FOR VARIOUS $K$ IN A $3 \times 3 \times 2\,\mathrm{m}^3$ ROOM

| $K$ | $i=1$ | $i=2$ | $i=3$ | $i=4$ | $i=5$ | $i=6$ | $i=7$ | $i=8$ | $i=9$ |
|---|---|---|---|---|---|---|---|---|---|
| 3 | (35.4, 140.1) | (33.3, 36.4) | (29.6, 262.7) | - | - | - | - | - | - |
| 4 | (38.9, 90.6) | (41.5, 0.10) | (41.8, 180.1) | (38.8, 270.2) | - | - | - | - | - |
| 5 | (0.1, 211.1) | (50.7, 180.0) | (50.4, 359.9) | (50.6, 90.0) | (50.6, 270.0) | - | - | - | - |
| 6 | (17.2, 306.9) | (54.5, 266.1) | (22.5, 140.4) | (52.2, 360.0) | (52.4, 84.0) | (55.8, 185.2) | - | - | - |
| 7 | (5.4, 305.9) | (53.8, 170.7) | (27.8, 43.8) | (58.9, 355.7) | (54.4, 96.5) | (35.1, 220.0) | (54.8, 278.6) | - | - |
| 8 | (6.5, 318.3) | (61.5, 268.3) | (27.3, 317.0) | (51.8, 89.4) | (57.8, 5.80) | (53.7, 171.3) | (38.0, 200.3) | (39.3, 91.1) | - |
| 9 | (2.50, 28.2) | (36.5, 266.8) | (33.9, 182.3) | (56.9, 178.7) | (42.2, 78.4) | (53.1, 97.5) | (57.9, 359.7) | (37.1, 355.1) | (58.2, 272.1) |

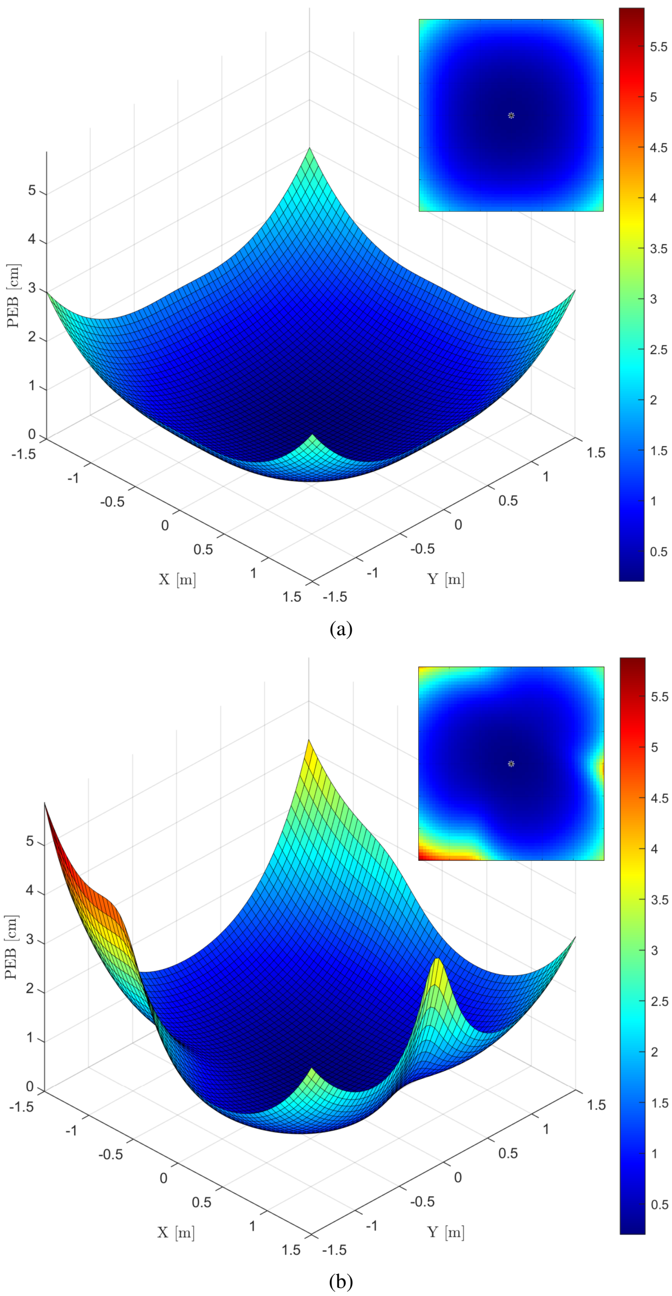


Fig. 3. Heat map of the PEB across a $3 \times 3\,\mathrm{m}$ testbed at $z = 0.8$ m for $K = 5$ orientations: (a) optimal set, (b) random set.

$\mathrm{PEB}_{90\%}$ decreases as $K$ increases, indicating improved localisation precision with additional orientations. A balance among acquisition latency, computational load, and positioning accuracy is attained at $K = 5$. For $\Phi_{1/2} = 30°$, at least four orientations are necessary due to the increased directivity limiting coverage with only three orientations; configurations with $K \geq 4$ outperform alternatives. In contrast, $\Phi_{1/2} = 45°$ achieves competitive error performance starting at $K = 3$, and given its widespread use in commercial LEDs, it is adopted for the subsequent simulations.

Figure 5 illustrates the dependence of the $\mathrm{PEB}_{90\%}$ on the SNR for various optimized $K$-orientation sets. As the SNR increases, $\mathrm{PEB}_{90\%}$ decreases monotonically. Moreover, at any fixed SNR, increasing $K$ yields a lower PEB, and the spacing between the curves diminishes at high SNR, indicating reduced incremental benefit from additional orientations under noise-limited conditions.

## V. NONLINEAR ESTIMATOR

The optical power collected by the PD depends nonlinearly on its relative position to the LED, through the range $d$, the irradiance angle $\phi_i$, and the incidence angle $\psi$. Denoting the receiver's orientation vector as $\mathbf{n}_r = [\alpha, \beta, \gamma]^\mathsf{T}$, we have

$$\cos\psi = \mathbf{n}_r \cdot (-\mathbf{n}_d) = -\frac{\alpha x + \beta y + \gamma\,(z - H)}{d}, \tag{24}$$

where $\mathbf{r} = (x, y, z)^\mathsf{T}$ is the receiver position and $H$ the LED height. Substituting (4), (24), and $d = \sqrt{x^2 + y^2 + (z - H)^2}$ into the received-power model (2), and then forming the sample mean as in (5), yields:

$$\bar{P}_r^{(i)} = -C\,\frac{\big(a_i x + b_i y + c_i (z - H)\big)^m}{\left[x^2 + y^2 + (z - H)^2\right]^{\frac{m+3}{2}}}(\alpha x + \beta y + \gamma (z - H)) + n, \tag{25}$$

where $n \sim \mathcal{N}(0, \sigma^2/N)$ and $C$ is defined in (8). Thus, the observations relate nonlinearly to the receiver coordinates $\mathbf{r}$. To recover the direction unit vector $\mathbf{n}_d$ rather than the full coordinate triple, one imposes the constraint $x^2 + y^2 + (z - H)^2 = 1$.

Defining:

$$Q_i(x, y, z) = a_i x + b_i y + c_i\,(z - H), \tag{26}$$

$$L(x, y, z) = \alpha x + \beta y + \gamma\,(z - H), \tag{27}$$

the model in (25) can be compactly written as:

$$\bar{P}_r^{(i)} = -\,C\,Q_i^m(x, y, z)\,L(x, y, z) + n. \tag{28}$$

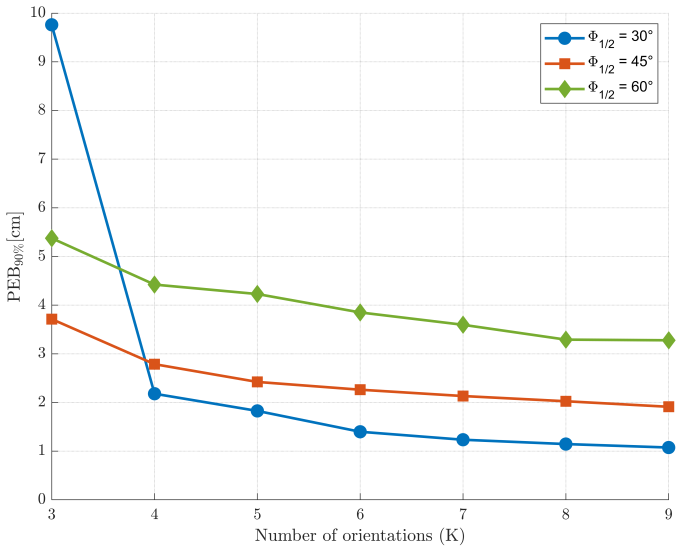


Fig. 4. $\text{PEB}_{90\%}$ versus number of orientations under three $\Phi_{1/2}$ scenarios with GA-optimised sets of orientations.

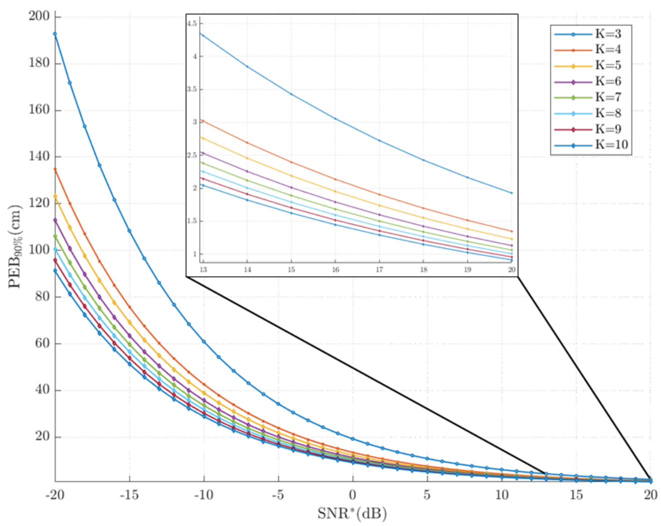


Fig. 5. Variation of the $\text{PEB}_{90\%}$ with $\overline{\text{SNR}}$ for different optimized $K$-orientation configurations.

This scenario exemplifies a classical estimation problem in which the averaged observations $\bar{P}_r^{(i)}$ for each orientation $i$ are nonlinearly related to the unknown receiver coordinates and corrupted by additive Gaussian noise of known variance. Although no unbiased efficient estimator exists for this NL model, one can derive a MVU estimate by solving a constrained NL least-squares problem.

Define the cost function:

$$F(x, y, z) = \sum_{i=1}^{K} F_i(x, y, z), \tag{29}$$

where:

$$F_i(x, y, z) = \left[\bar{P}_r^{(i)} + C\, Q_i(x, y, z)\, L(x, y, z)\right]^2. \tag{30}$$

Here $Q_i$ and $L$ are as defined in (26)–(27), and $C$ is the optical constant introduced in (7).

Since our goal is to recover the unit-norm direction vector $\mathbf{n}_d$ rather than the absolute coordinates, the variables $\big(x, y, z - H\big)$ must lie on the unit sphere $\mathbb{S}^2$. Moreover, the polynomial terms $Q_i(x, y, z)$ and $L(x, y, z)$ must satisfy:

$$Q_i(x, y, z) \geq 0, \quad L(x, y, z) \leq 0, \tag{31}$$

corresponding respectively to $\cos\phi_i \geq 0$ and $\cos\psi \geq 0$ for incidence and irradiance angles in $[0°, 90°]$.

Thus, the NL direction-finding estimator is:

$$\hat{\mathbf{n}}_{d,\text{NL}} = \arg \min_{\substack{(x,y,z-H)\in\mathbb{S}^2 \\ Q_i(x,y,z)\geq 0 \\ L(x,y,z)\leq 0}} F(x, y, z), \tag{32}$$

with $F(x, y, z)$ given by (29)–(30). In practice, this optimisation can be performed in MATLAB using the `solve` function, which internally invokes `fmincon` and employs gradient-descent algorithms suitable for constrained NL problems.

## VI. Linear Estimator

The present section develops the direction-finding stage with full mathematical rigour and presents two statistically grounded estimators: (i) generalised least squares (GLS), which is minimum-variance under Gaussian noise with known covariance, and (ii) weighted least squares (WLS), an inexpensive approximation that neglects the weak cross–correlation between equations. For implementation details, an end-to-end pseudocode of the full pipeline (GLS/WLS direction finding, distance recovery, and 3D position estimation) is provided in Algorithm 1.

### A. Direction Estimation via GLS

This subsection consolidates the analytical procedure that converts the raw optical powers into a closed-form, statistically efficient estimate of the direction vector $\mathbf{n}_d$. The exposition proceeds from the deterministic constraints, through a rigorous stochastic characterisation of the residuals, to the final GLS solution.

*1) Lambertian power ratios and homogeneous constraints:* Let $\mu_i = \mathbb{E}\big[P_r^{(i)}\big]$ be the noise-free power at orientation $i$. We define power ratio $\beta$ as:

$$\beta_i = \big(\mu_i/\mu_1\big)^{1/m}, \quad i = 2, \ldots, K, \tag{33}$$

and, from (2), (3), (4) and (8):

$$\mu_i = -C\, \frac{\big(\mathbf{n_t^{(i)}} \cdot \mathbf{d}\big)^m \big(\mathbf{n_r} \cdot \mathbf{d}\big)}{d^{m+3}}, \tag{34}$$

using (34) in (33), each $\beta_i$ imposes the orthogonality condition:

$$\mathbf{a}_i \cdot \mathbf{d} = 0, \quad i = 2, \ldots, K, \tag{35}$$

i.e., $\mathbf{a}_i \perp \mathbf{d}$, with:

$$\mathbf{a}_i = \mathbf{n}_t^{(i)} - \beta_i \mathbf{n}_t^{(1)}, \tag{36}$$

therefore supplies $K-1$ homogeneous hyperplanes that intersect at the true $\mathbf{d}$.

*2) Stochastic perturbation of the ratios:* Recall from (5) that the sample mean is:

$$\bar{P}_r^{(i)} = \frac{1}{N}\sum_k P_{r,k}^{(i)} \sim \mathcal{N}(\mu_i, \sigma^2/N). \tag{37}$$

For notational convenience, define $\hat{\mu}_i \triangleq \bar{P}_r^{(i)}$, which is the MVU estimator of $\mu_i$. Injecting $\hat{\mu}_i$ into (33) yields:

$$\hat{\beta}_i = g_i(\hat{\mu}_1, \hat{\mu}_i), \qquad g_i(x_1, x_2) = \left(\frac{x_2}{x_1}\right)^{\frac{1}{m}}. \tag{38}$$

A first-order Taylor expansion of $g_i$ centred at $\boldsymbol{\mu}_i = (\mu_1, \mu_i)^\mathsf{T}$ gives:

$$\hat{\beta}_i \approx \beta_i + \frac{\beta_i}{m}\left(\frac{n_i}{\mu_i} - \frac{n_1}{\mu_1}\right), \tag{39}$$

Hence every $\hat{\beta}_i$ shares the common perturbation $n_1$, producing correlation across orientations.

Define the ratio error:

$$\tilde{n}_i = \hat{\beta}_i - \beta_i. \tag{40}$$

Using (39) and the independence of $n_i$ and $n_1$,

$$\mathrm{Var}[\tilde{n}_i] = \frac{\sigma^2}{Nm^2}\,\beta_i^2\big(\mu_i^{-2} + \mu_1^{-2}\big), \tag{41}$$

$$\mathrm{Cov}[\tilde{n}_i, \tilde{n}_j] = \frac{\sigma^2}{Nm^2}\,\beta_i\beta_j\,\mu_1^{-2}, \quad i \neq j. \tag{42}$$

The factor $(\mu_i^{-2} + \mu_1^{-2})$ in (41) demonstrates that the variance of $\tilde{n}_i$ depends explicitly on $\mu_i$, even though each $n_i$ has the same variance $\sigma^2$. In other words, a single-LED system exhibits *heteroscedastic* ratio errors.

*3) Residuals and their covariance:* Define the empirical constraint normals as $\hat{\mathbf{a}}_i = \mathbf{n}_t^{(i)} - \hat{\beta}_i\,\mathbf{n}_t^{(1)}$ and the corresponding residuals as:

$$\xi_i = \hat{\mathbf{a}}_i \cdot \mathbf{d}, \qquad i = 2, \ldots, K. \tag{43}$$

Using (40) and (36), we have $\hat{\mathbf{a}}_i = \mathbf{a}_i - \tilde{n}_i\,\mathbf{n}_t^{(1)}$. Since $\mathbf{a}_i \cdot \mathbf{d} = 0$ by (35), it follows that:

$$\xi_i = -\big(\mathbf{n}_t^{(1)} \cdot \mathbf{d}\big)\,\tilde{n}_i = -\,\kappa\,\tilde{n}_i, \qquad \kappa = \mathbf{n}_t^{(1)} \cdot \mathbf{d}. \tag{44}$$

Stacking $\boldsymbol{\xi} = [\xi_2, \ldots, \xi_K]^\mathsf{T}$ and $\tilde{\boldsymbol{n}} = [\tilde{n}_2, \ldots, \tilde{n}_K]^\mathsf{T}$ yields:

$$\begin{aligned} \boldsymbol{\xi} &= -\,\kappa\,\tilde{\boldsymbol{n}}, \\ \mathbb{E}[\boldsymbol{\xi}] &= \mathbf{0}, \\ \boldsymbol{\Sigma}_{\boldsymbol{\xi}} &= \mathrm{Cov}(\boldsymbol{\xi}) = \kappa^2\boldsymbol{\Sigma}_\beta, \end{aligned} \tag{45}$$

where $\boldsymbol{\Sigma}_\beta \in \mathbb{R}^{(K-1)\times(K-1)}$ collects the variances and covariances of $\tilde{n}_i$ given in (41)–(42).

*4) GLS formulation and closed-form solution:* Stack the empirical constraint normals into $\hat{\mathbf{A}} = [\hat{\mathbf{a}}_2 \cdots \hat{\mathbf{a}}_K] \in \mathbb{R}^{3\times(K-1)}$. From (43)–(45), the stacked residuals satisfy $\boldsymbol{\xi} = \hat{\mathbf{A}}^\mathsf{T}\mathbf{d}$ with $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{\xi}})$ and $\boldsymbol{\Sigma}_{\boldsymbol{\xi}} = \kappa^2\boldsymbol{\Sigma}_\beta$, $\kappa = \mathbf{n}_t^{(1)} \cdot \mathbf{d}$. Given an observation of $\boldsymbol{\xi} = \hat{\mathbf{A}}^\mathsf{T}\mathbf{d}$, the maximum-likelihood (equivalently, GLS) estimate of the direction is obtained by minimizing the Mahalanobis norm of $\boldsymbol{\xi}$:

$$\begin{aligned} \hat{\mathbf{d}} &= \arg\min_{\|\mathbf{d}\|=1} \big\|\hat{\mathbf{A}}^\mathsf{T}\mathbf{d}\big\|^2_{\boldsymbol{\Sigma}_{\boldsymbol{\xi}}^{-1}} \\ &= \arg\min_{\|\mathbf{d}\|=1} \mathbf{d}^\mathsf{T}\big(\hat{\mathbf{A}}\,\boldsymbol{\Sigma}_{\boldsymbol{\xi}}^{-1}\hat{\mathbf{A}}^\mathsf{T}\big)\mathbf{d}. \end{aligned} \tag{46}$$

**Algorithm 1** 3D Position Estimation from $n$ Orientations

**Require:** Power samples $P_{r,k}^{(i)}$ $(k = 1, \ldots, N,\ i = 1, \ldots, K)$;
steering vectors $\mathbf{n}_t^{(i)}$; Lambertian order $m$;
noise variance $\sigma^2$; MODE $\in \{\mathrm{GLS}, \mathrm{WLS}\}$;
radiometric constant $C$.
**Ensure:** $\hat{\mathbf{r}}$ (3D position of the receiver)

**Direction finding:**
  Compute $\hat{\mu}_i = \frac{1}{N}\sum_{k=1}^{N} P_{r,k}^{(i)}$ for $i = 1, \ldots, K$
  Compute $\hat{\beta}_i = (\hat{\mu}_i/\hat{\mu}_1)^{1/m}$ for $i = 2, \ldots, K$
  Form $\mathbf{a}_i = \mathbf{n}_t^{(i)} - \hat{\beta}_i\,\mathbf{n}_t^{(1)}$
  Set $\mathbf{A} = [\mathbf{a}_2, \ldots, \mathbf{a}_n]$
**if** MODE = GLS **then**
    Build $\boldsymbol{\Sigma}_\beta$ via (41)–(42), substituting $\beta_i \leftarrow \hat{\beta}_i$
    $\mathbf{M} \leftarrow \mathbf{A}\,\boldsymbol{\Sigma}_\beta^{-1}\,\mathbf{A}^\mathsf{T}$
**else** ▷ WLS (diagonal approximation)
    $\tilde{\boldsymbol{\Sigma}}_\beta \leftarrow \mathrm{diag}\big\{\hat{\beta}_i^2\big(\hat{\mu}_i^{-2} + \hat{\mu}_1^{-2}\big)\big\}$
    $\mathbf{M} \leftarrow \mathbf{A}\,\tilde{\boldsymbol{\Sigma}}_\beta^{-1}\,\mathbf{A}^\mathsf{T}$
**end if**
  Compute $\mathbf{v}_{\min}$ = eigenvector of $M$ with smallest eigenvalue
  $\hat{\mathbf{n}}_d \leftarrow \mathbf{v}_{\min}/\|\mathbf{v}_{\min}\|$
**if** $\hat{\mathbf{n}}_d \cdot \mathbf{n}_t^{(1)} < 0$ **then**
    $\hat{\mathbf{n}}_d \leftarrow -\,\hat{\mathbf{n}}_d$ ▷ Ensure pointing to Rx
**end if**

**Distance Estimation (beamsteering):**
  Set $\mathbf{n}_t \leftarrow \hat{\mathbf{n}}_d$, $\mathbf{n}_r \leftarrow -\,\hat{\mathbf{n}}_d$
  Collect $N$ power samples $P_r^{*(k)}$
  Compute $\hat{\mu}^* = \frac{1}{N}\sum_{k=1}^{N} P_{r,k}^*$
  $\hat{d} \leftarrow \sqrt{\frac{C}{\hat{\mu}^*}}$

**Final 3D position:**
  $\hat{\mathbf{r}} \leftarrow \mathbf{t} + \hat{d}\,\hat{\mathbf{n}}_d$
**return** $\hat{\mathbf{r}}$

Since $\boldsymbol{\Sigma}_{\boldsymbol{\xi}} = \kappa^2\boldsymbol{\Sigma}_\beta$, the unknown scale $\kappa^2$ cancels in (46); hence one may use $\boldsymbol{\Sigma}_\beta^{-1}$ in place of $\boldsymbol{\Sigma}_{\boldsymbol{\xi}}^{-1}$ without affecting the minimizer. The problem is a constrained Rayleigh quotient whose solution is the unit eigenvector associated with the smallest eigenvalue of:

$$\mathbf{M}_{\mathrm{GLS}} = \hat{\mathbf{A}}\,\boldsymbol{\Sigma}_\beta^{-1}\hat{\mathbf{A}}^\mathsf{T} \in \mathbb{R}^{3\times 3}. \tag{47}$$

Therefore,

$$\hat{\mathbf{n}}_{d,\mathrm{GLS}} = \frac{\mathbf{v}_{\min}\big(\mathbf{M}_{\mathrm{GLS}}\big)}{\big\|\mathbf{v}_{\min}\big(\mathbf{M}_{\mathrm{GLS}}\big)\big\|}, \qquad \hat{\mathbf{n}}_{d,\mathrm{GLS}} \cdot \mathbf{n}_t^{(1)} > 0, \tag{48}$$

where $\mathbf{v}_{\min}(\cdot)$ denotes the eigenvector corresponding to the smallest eigenvalue; the sign convention ensures the direction points from the transmitter toward the receiver.

### *B. WLS as a practical simplification*

The GLS solution in (46) requires inverting the $(K-1)\times(K-1)$ covariance $\boldsymbol{\Sigma}_{\boldsymbol{\xi}} = \kappa^2\boldsymbol{\Sigma}_\beta$, typically via a Cholesky factorization [41]. For real-time, low-power receivers, a lighter alternative is to neglect the off-diagonal terms in (42) and approximate:

$$\tilde{\boldsymbol{\Sigma}}_\beta = \mathrm{diag}\Big\{\beta_i^2\big(\mu_i^{-2} + \mu_1^{-2}\big)\Big\}_{i=2}^{K}. \tag{49}$$

Since $\boldsymbol{\Sigma}_{\boldsymbol{\xi}} = \kappa^2\boldsymbol{\Sigma}_\beta$, the unknown scale $\kappa^2$ cancels, and we define the WLS weighting matrix as $\mathbf{W} = \tilde{\boldsymbol{\Sigma}}_\beta^{-1}$.

Using the empirical normals $\hat{\mathbf{A}}$ from the previous subsection, the WLS criterion is the Rayleigh quotient:

$$J_{\mathrm{WLS}}(\mathbf{d}) = \mathbf{d}^{\mathsf{T}}(\hat{\mathbf{A}}\,\mathbf{W}\,\hat{\mathbf{A}}^{\mathsf{T}})\mathbf{d}, \qquad \|\mathbf{d}\| = 1, \tag{50}$$

and we introduce the associated WLS matrix:

$$\mathbf{M}_{\mathrm{WLS}} = \hat{\mathbf{A}}\,\mathbf{W}\,\hat{\mathbf{A}}^{\mathsf{T}} \in \mathbb{R}^{3\times 3}. \tag{51}$$

The minimizer is the unit eigenvector corresponding to the smallest eigenvalue of $\mathbf{M}_{\mathrm{WLS}}$:

$$\hat{\mathbf{n}}_{d,\mathrm{WLS}} = \frac{\mathbf{v}_{\min}(\mathbf{M}_{\mathrm{WLS}})}{\left\|\mathbf{v}_{\min}(\mathbf{M}_{\mathrm{WLS}})\right\|}, \qquad \hat{\mathbf{n}}_{d,\mathrm{WLS}} \cdot \mathbf{n}_t^{(1)} > 0. \tag{52}$$

The neglected cross-covariances in (42) scale with $\mu_1^{-2}$, whereas the retained diagonal terms scale with $\mu_i^{-2} + \mu_1^{-2}$. Thus, when the reference orientation $i = 1$ is chosen such that $\mu_1 \gg \mu_i$ for most $i$, the off-diagonal terms are negligible relative to the diagonal terms and WLS closely tracks GLS while avoiding a full covariance inversion. In more symmetric geometries where $\mu_i \approx \mu_1$ for many $i$, the cross-correlations are non-negligible and the full GLS should be used for optimal performance.

## VII. Results

This section compares the model-based estimators introduced earlier (i.e., NL, WLS, GLS). Figure 6 plots the cumulative distribution function (CDF) of the 3D position error for $K \in \{3, 5, 9\}$. The case $K = 3$ uses the deterministic baseline of [22] (i.e., no WLS/GLS weighting), $K = 5$ is selected as a latency–accuracy compromise based on the CRLB analysis, and $K = 9$ represents the upper bound of the orientation counts studied in this paper. For $K = 3$, $K = 5$ (GLS, WLS), and $K = 9$ (GLS, WLS), the GA-optimized orientation sets from Table III were used.

The NL estimator exhibited degraded performance when paired with those $K$-optimal sets because it minimizes (32) without an explicit weighting matrix, thereby giving comparable influence to orientations with negligible received power (e.g., $\phi_i > 90°$), which impairs convergence. To provide a fair comparison, a separate GA optimization was conducted using the parameters in Table II but with the NL method's RMSE as the objective; this NL-specific optimal set is used in the NL curves reported.

Across all estimators, the $K = 9$ configurations (dashed traces) stochastically dominate their $K = 5$ counterparts, in agreement with the CRLB trend that additional orientations reduce the PEB. Among the model-based schemes, GLS consistently tracked the theoretical PEB most closely, regardless of $K$, with WLS incurring a small but visible loss due to its diagonal covariance approximation. All proposed estimators substantially outperformed the deterministic $K = 3$ baseline [22].

Table IV summarizes the quantitative metrics: RMSE, the 90th percentile of the error CDF ($\mathrm{CDF}_{90\%}$), and average positioning error (APE). For $K = 5$, GLS achieved $\mathrm{RMSE} =$ 2.72 cm and $\mathrm{CDF}_{90\%} = 3.86$ cm, within a factor $\approx 1.75\times$ of the CRLB benchmark (1.55 cm RMSE). WLS trailed GLS by $\approx 36\%$ in RMSE at $K = 5$, reflecting the benefit of modeling

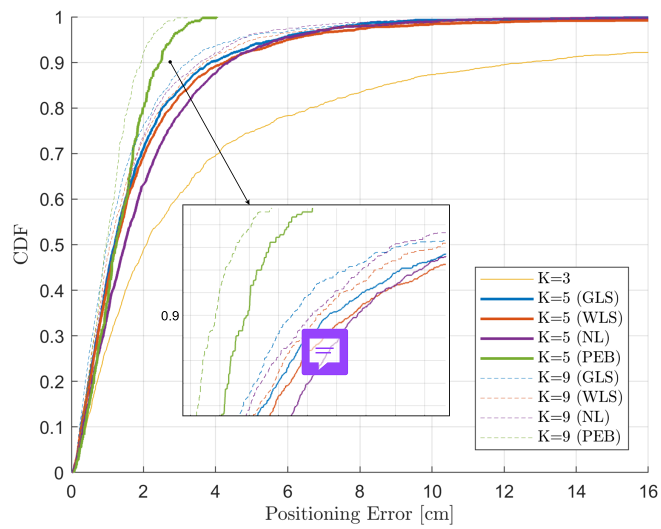


Fig. 6. Comparison of the CDFs of different 3D position estimation methods for $K = 3, 5, 9$.

TABLE IV
Performance of Position-Estimation Methods

| $K$ | Method | RMSE [cm] | CDF 90% [cm] | APE [cm] |
|---|---|---|---|---|
| 3 | [22] | 11.05 | 12.61 | 5.35 |
| 5 | GLS | 2.72 | 3.86 | 1.79 |
| 5 | WLS | 3.71 | 4.24 | 2.02 |
| 5 | NL | 2.90 | 4.35 | 2.05 |
| 5 | CRLB | 1.55 | 2.48 | 1.34 |
| 9 | GLS | 2.41 | 3.23 | 1.54 |
| 9 | WLS | 2.96 | 3.67 | 1.75 |
| 9 | NL | 2.36 | 3.57 | 1.64 |
| 9 | CRLB | 1.21 | 1.93 | 1.04 |

inter-orientation error correlations. The NL estimator, after re-optimizing its orientation set, became competitive with WLS at $K = 5$ and surpassed WLS at $K = 9$, where NL attained an RMSE of 2.36 cm versus 2.96 cm for WLS. Increasing $K$ from 5 to 9 delivered consistent gains for all methods (e.g., GLS RMSE improved from 2.72 cm to 2.41 cm), and the gap to the CRLB narrowed (from 1.55 cm to 1.21 cm). Overall, GLS provided the best fidelity to the bound with modest complexity; WLS offered a favorable accuracy–complexity compromise; and NL benefited markedly from orientation-set co-design, especially at larger $K$. The APE values track the same ordering as RMSE and $\mathrm{CDF}_{90\%}$, indicating that the improvements are uniform across the distribution rather than confined to the median or the tail.

Figure 7 reports the end-to-end computation latency to produce a 3D position estimate once the powers are available. The NL method exhibits the largest latency (mean $\approx 92\,\mathrm{ms}$) due to its iterative constrained optimization. In contrast, GLS and WLS reduce inference to a single eigenproblem, yielding mean latencies of $\approx 49\,\mu$s and $\approx 41\,\mu$s, respectively. These results indicate that closed-form WLS/GLS enables real-time operation without fingerprinting or model retraining.

Figure 8 illustrates the 3D position estimates produced by GLS and WLS for $K = 5$ at three receiver heights $z \in \{0,\, 0.6,\, 1.2\}\,\mathrm{m}$, overlaid on the ground truth. At $z = 1.2\,\mathrm{m}$,

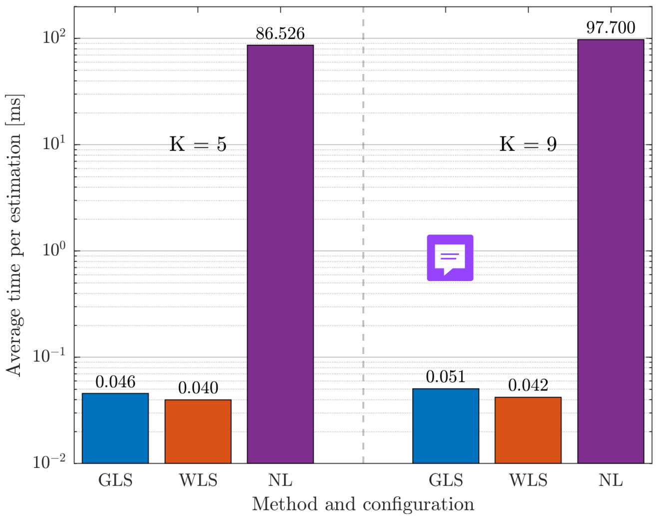


Fig. 7. Computation latency of the NL, GLS, and WLS estimators for $K = 5$ and $K = 9$.

errors increase near the testbed boundaries—particularly at the corners—consistent with the PEB heat maps (Fig. 3a) and attributable to reduced SNR at extreme incidence angles. At $z = 0\,\mathrm{m}$, the errors are more spatially uniform. In both cases, GLS maintains a tighter scatter than WLS, in line with its closer adherence to the CRLB.

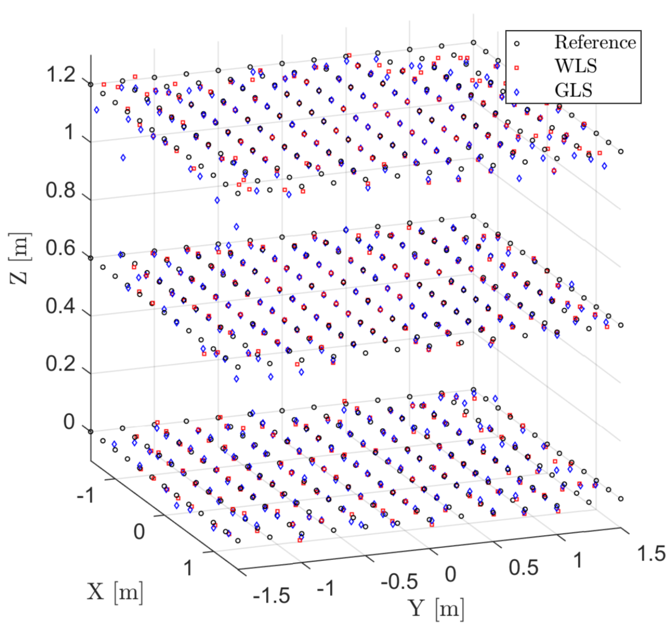


Fig. 8. 3D position estimates at three receiver heights for $K = 5$ orientations: GLS, WLS, ground truth.

## VIII. Conclusion

This paper introduced a beam–steered single-LED, single-PD OWP architecture that delivers full 3D indoor localization without receiver rotation, cameras, or PD arrays. The method is explicitly model-based, grounded in radiometry and geometry, so it requires no fingerprinting or data retraining, enabling interpretable performance, principled uncertainty analysis, and efficient co-design of hardware and algorithms.

We proposed a two-stage pipeline: (i) direction finding from $K$ steered orientations, followed by (ii) range recovery from a single beam-aligned power measurement. A full CRLB/PEB analysis characterized identifiability ($K \geq 3$) and showed the benefits of overdetermination ($K \geq 4$). A GA-based orientation-set optimizer minimized the RMSE PEB across a 3D testbed and yielded practical steering patterns. Additionally, we introduce a ratio-based linearization of the Lambertian power model that yields closed-form GLS/WLS direction estimators for stage (i).

Simulations demonstrated centimeter-level accuracy with modest $K$. Optimized orientation sets achieved sub–1.5 cm median PEB across $K \in [3, 9]$, reaching $0.74\mathrm{cm}$ at $K = 9$. With $K = 5$, GLS attained RMSE $2.72\mathrm{cm}$ and $\mathrm{CDF}_{90\%} = 3.86\mathrm{cm}$, tracking the CRLB within a factor $\approx 1.75\times$; increasing to $K = 9$ improved GLS to RMSE $2.41\mathrm{cm}$. Importantly, GLS/WLS reduce inference to a single $3 \times 3$ eigenproblem, yielding $\mu$s-level latency, while the nonlinear baseline exhibits ms-level runtimes. These results confirm that closed-form, model-based estimation can closely approach the theoretical bound at real-time speeds.

Beyond accuracy, the architecture is deployment-friendly. By concentrating complexity in the infrastructure (beam steering at a single luminaire) and using only one PD on the user side, the system avoids dense LED constellations and specialized receivers. This modular, single-transmitter design can time-multiplex its beam to serve multiple users, enabling position estimates for an $N$-robot swarm with minimal on-robot implementation.

## References

[1] A. Sesyuk, S. Ioannou, and M. Raspopoulos, “A survey of 3d indoor localization systems and technologies,” 12 2022.
[2] S. He and S. H. Chan, “Wi-fi fingerprint-based indoor positioning: Recent advances and comparisons,” pp. 466–490, 1 2016.
[3] R. Faragher and R. Harle, “Location fingerprinting with bluetooth low energy beacons,” *IEEE Journal on Selected Areas in Communications*, vol. 33, pp. 2418–2428, 11 2015.
[4] A. Alarifi *et al.*, “Ultra wideband indoor positioning technologies: Analysis and recent advances,” 5 2016.
[5] L. M. Ni, Y. Liu, Y. C. Lau, and A. P. Patil, “Landmarc: Indoor location sensing using active rfid,” Tech. Rep., 2004.
[6] L. Italiano *et al.*, “A tutorial on 5g positioning,” pp. 1488–1535, 6 2025.
[7] F. Zafari, A. Gkelias, and K. K. Leung, “A survey of indoor localization systems and technologies,” *IEEE Communications Surveys and Tutorials*, vol. 21, pp. 2568–2599, 2019.
[8] D. Lymberopoulos *et al.*, “A realistic evaluation and comparison of indoor location technologies: Experiences and lessons learned,” in *IPSN 2015 - Proceedings of the 14th International Symposium on Information Processing in Sensor Networks (Part of CPS Week)*. United States: Association for Computing Machinery, Apr. 2015, pp. 178–189.
[9] S. G. Leitch *et al.*, “On indoor localization using wifi, ble, uwb, and imu technologies,” *Sensors (Basel, Switzerland)*, vol. 23, 10 2023.
[10] R. Wang, G. Niu, Q. Cao, C. S. Chen, and S.-W. Ho, “A survey of visible-light-communication-based indoor positioning systems,” *Sensors*, vol. 24, pp. 164–169, 8 2024.
[11] S. E. Trevlakis *et al.*, “Localization as a key enabler of 6g wireless systems: A comprehensive survey and an outlook,” *IEEE Open Journal of the Communications Society*, vol. 4, pp. 2733–2801, 2023.
[12] H. Haas, L. Yin, Y. Wang, and C. Chen, “What is lifi?” *J. Lightwave Technol.*, vol. 34, no. 6, pp. 1533–1544, Mar 2016.
[13] V. P. Rekkas *et al.*, “Artificial intelligence in visible light positioning for indoor iot: A methodological review,” *IEEE Open Journal of the Communications Society*, vol. 4, pp. 2838–2869, 2023.

[14] H. Q. Tran and C. Ha, "Machine learning in indoor visible light positioning systems: A review," *Neurocomputing*, vol. 491, pp. 117–131, 6 2022.

[15] T. Glass, F. Alam, M. Legg, and F. Noble, "Autonomous fingerprinting and large experimental data set for visible light positioning," *Sensors*, vol. 21, 5 2021.

[16] L. Deng, Y. Fan, Q. Zhao, and P. Wu, "Indoor visible light positioning based on channel estimation and cramér-rao low bound analysis with random receiving orientation of user equipment," 2023.

[17] L. Qin, B. Niu, B. Li, and Y. Du, "Indoor visible light high precision three-dimensional positioning algorithm based on single led lamp," *Optik*, vol. 207, 4 2020.

[18] H. Chen *et al.*, "High accuracy indoor visible light positioning using a long short term memory-fully connected network based algorithm," *Optics Express*, vol. 29, p. 41109, 12 2021.

[19] L. Qin, B. Niu, B. S. Li, X. L. Hu, and Y. X. Du, "High precision indoor positioning algorithm of single led lamp based on a-bayes," *Optik*, vol. 241, 9 2021.

[20] R. Liu, Z. Liang, K. Yang, and W. Li, "Machine learning based visible light indoor positioning with single-led and single rotatable photo detector," *IEEE Photonics Journal*, vol. 14, 6 2022.

[21] R. Zia-Ul-Mustafa, O. I. Younus, H. L. Minh, Z. Ghassemlooy, and S. Zvanovec, "A single led-based indoor visible light positioning system-recent trends and the impact of ambient light on positioning accuracy," in *2023 South American Conference on Visible Light Communications, SACVLC 2023*. Institute of Electrical and Electronics Engineers Inc., 2023, pp. 112–117.

[22] L. Chassagne *et al.*, "Optical wireless positioning by modulation of the optical source orientation," *Optics Express*, vol. 33, p. 13161, 3 2025.

[23] J. Zhang and X. Ke, "Delineating regional bes–elm neural networks for studying indoor visible light positioning," *Photonics*, vol. 11, 10 2024.

[24] S. Li, L. Qin, D. Zhao, and X. Hu, "Indoor positioning system for single led light based on deep residual shrinkage network," *Optics Communications*, vol. 559, 5 2024.

[25] Z. Wang, Z. Liang, R. Liu, X. Li, and H. Li, "Design and performance analysis for indoor visible light positioning with single led and single-tilted-rotatable pd," *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1–14, 2024.

[26] S. Ma *et al.*, "Centimeter-level 3-d mobile online visible light positioning system with single led lamp," *IEEE Internet of Things Journal*, vol. 11, pp. 418–429, 1 2024.

[27] H. Shi, C. Wang, and Y. Zhao, "Visible light positioning using a single led and a single rotatable photodetector," *IEEE Photonics Technology Letters*, 2025.

[28] J. Shi *et al.*, "Single led visible light positioning scheme based on intelligent reflecting surfaces," *Applied Optics*, vol. 64, p. 4438, 5 2025.

[29] Y. Zhuang *et al.*, "A survey of positioning systems using visible led lights," *IEEE Communications Surveys and Tutorials*, vol. 20, pp. 1963–1988, 2018.

[30] Plets, David and Almadani, Yousef and Bastiaens, Sander and Ijaz, Muhammad and Martens, Luc and Joseph, Wout, "Efficient 3D trilateration algorithm for visible light positioning," *JOURNAL OF OPTICS*, vol. 21, no. 5, p. 4, 2019.

[31] Y. Almadani *et al.*, "A novel 3d visible light positioning method using received signal strength for industrial applications," *Electronics*, vol. 8, no. 11, 2019.

[32] Y. Almadani, M. Ijaz, B. Adebisi, S. Rajbhandari, and S. Bastiaens, "An experimental evaluation of a 3d visible light positioning system in an industrial environment with receiver tilt and multipath reflections," *Optics Communications*, vol. 483, p. 126654, 2021.

[33] Z. Zhang *et al.*, "Demonstration of three-dimensional indoor visible light positioning with multiple photodiodes and reinforcement learning," *Sensors (Switzerland)*, vol. 20, pp. 1–14, 11 2020.

[34] G. Cossu and E. Ciaramella, "A 3d indoor positioning system based on common visible leds," *Physical Communication*, vol. 54, 10 2022.

[35] F. Wu, N. Stevens, L. D. Strycker, and F. Rottenberg, "Enhancing rss-based visible light positioning by optimal calibration of led tilt and gain," *IEEE Transactions on Communications*, 2025.

[36] W. Liu *et al.*, "A real-time mems mirror based beam steering and tracking system for optical wireless communication," *J. Lightwave Technol.*, vol. 43, no. 6, pp. 2681–2689, Mar 2025.

[37] F. Feng, P. Sangwongngam, G. Faulkner, and D. O'Brien, "Wide field-of-view optical broadcasting for bi-directional indoor optical wireless communications employing pam-4 modulation," *Optics Letters*, vol. 44, pp. 6009–6012, 12 2019.

[38] C.-Y. Hsu, G.-Z. Yiu, and Y.-C. Chang, "Free-space applications of silicon photonics: A review," *Micromachines*, vol. 13, no. 7, 2022.

[39] J. Kahn and J. Barry, "Wireless infrared communications," *Proceedings of the IEEE*, vol. 85, no. 2, pp. 265–298, 1997.

[40] S. M. Kay, *Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory*. Englewood Cliffs, NJ: Prentice Hall, 1993.

[41] G. Golub and C. Van Loan, *Matrix Computations*, ser. Johns Hopkins Studies in the Mathematical Sciences. Johns Hopkins University Press, 2013.

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